2.5 Conditionals and Biconditionals

Lesson Introduction

<table>
<tr><td>Benchmark</td><td>MA.912.LT.4.3 Identify and accurately interpret “if…then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.</td><td>Learning Targets</td><td>I can write the converse, inverse, and contrapositive of a mathematical statement.
I can determine if a statement is biconditional.
I can write a definition as a biconditional.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td>Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td>N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="3">How can I determine if a statement is biconditional?
How can I determine if a statement written in “if…then” form is true?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="3">In this lesson, students continue their work with conditional, converse, inverse, and contrapositive statements. Students will engage in an activity where they determine the truth of the converse, inverse, and contrapositive of a given true conditional statement. They will then take what they observed in the activity to develop an understanding of a biconditional statement and its relationship to definitions. Biconditional statements will be written in “if and only if” form.</td></tr>
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Component Summary				
	2.5.1 Warm-Up (~5 min.)	Calculating music streaming payouts *(SE p. 102)*	2.5.3 Bring It Together (10 min.)	Connecting true conditionals and converses to biconditional statements and definitions *(SE p. 104)*
	2.5.2 Activity (20 min.)	Determining the truth value of conditionals, converses, inverses, and contrapositives *(SE p. 103)*	2.5.4 Your Turn! (10 min.)	Writing and determining the truth value of statements to determine if a biconditional exists *(SE p. 105)*
	2.5.5 Wrap-Up (~5 min.)	Self-assessment on student confidence with each concept from this lesson *(SE p. 106)*		

Resources

Glossary	Materials	Additional Preparation
Key Terms: • Biconditional • Definition Additional Terms: • Converse • Inverse • Contrapositive • Truth Value	N/A	• Work each of the problems in 2.5.2 Activity prior to the lesson. Determine if you may need to provide additional background knowledge to support the claims made in the conditional statements, as some students may not be familiar with these facts.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may write the inverse statement as the converse or vice-versa.
Students may forget to switch the hypothesis, p, and the conclusion, q, when writing a contrapositive statement.
Students may mistake the definition of biconditional and forget to confirm the converse is also true.
Students may assume that statements written as conditionals are always true.</td><td>Pacing will be a key factor in this lesson, especially with the 2.5.2 Activity. This activity has 10 conditional statements, each with a converse, inverse, and contrapositive truth value. Work through each problem and time yourself to determine whether you want all students to do all 10 or if you will need to divide them up between groups.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematic Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Poll the Class
2.5.2 Activity provides students with the opportunity to determine if the converse, inverse, and contrapositive of a conditional statement are true or false. Upon completion of each statement, consider polling the class to determine whether they believe it is true or false.</td><td>MA.K12.MTR.6.1: Reasonableness
2.5.2 Activity has students determine whether the converse, inverse, and contrapositive of a conditional statement are true or false. Students are given various true conditional statements grounded in geometry. Students then evaluate the resulting statement to determine its truth. The activity not only serves to build understanding of converse, inverse, and contrapositive of a conditional statement but also builds a student’s understanding of geometric concepts that then leads to understanding a biconditional.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson is heavy on new vocabulary and continues using the vocabulary: converse, inverse, and contrapositive statements. These can be easy for students to get mixed up. Consider providing a Frayer Model template to allow students to have a reference for future use.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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2.5.1 Warm-Up: Streaming Payouts

Component Overview: *Students will analyze an image that shows streaming payouts per platform. They will use this information to answer questions on how much a payout will be for a number of streams and how many streams are needed for a particular payout.*

2.5.1 Warm-Up: Streaming Payouts

The payouts to a musician for each time a song is streamed on multiple streaming platforms in 2019 is shown.

Platform	Payout Per Stream
Napster	$0.01900
Tidal	$0.01250
Apple Music	$0.00735
Google Play	$0.00676
Deezer	$0.00640
Spotify	$0.00437
Amazon	$0.00402
Pandora	$0.00133
YouTube	$0.00069

1. Determine the payout for a musician whose song was streamed 100,000 times on each platform.

Platform	Payout for 100,000 Streams (in dollars)
Napster	$1,900
Tidal	$1,250
Apple Music	$735
Google Play	$676
Deezer	$640
Spotify	$437
Amazon	$402
Pandora	$133
YouTube	$69

2. How many streams on Spotify did it take to for a musician to make $1 in 2019?
   229 *streams*

If time permits, allow students to share strategies they used to complete the table. Highlight the strategy of using a power of ten to quickly determine the payouts.

2.5.2 Activity: True or False

Component Overview: *Students will complete an activity where they are given a conditional statement and must determine whether or not its converse, inverse, and contrapositive statements are true or false. For this activity, students can work independently or with a partner or small group.*

2.5.2 Activity: True or False

For each conditional statement given, consider whether its converse, inverse, and contrapositive statements are true or false. Record the results in the table for each conditional.

1. **Conditional**: If all of the angles of a triangle are less than $90°$, then the triangle is an acute triangle.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

2. **Conditional**: If the sum of two angles is $90°$, then the angles are complementary.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

3. **Conditional**: If a polygon has six sides, then it is a hexagon.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

Timing will be a key consideration when doing this activity.
This activity has 9 conditional statements, each with a converse, inverse, and contrapositive truth value. Work through each problem and time yourself to determine whether you want all students to do all 9 or divide them between groups.
Because each conditional is a definition the converse, inverse, and contrapositive statements are true.
Students may second-guess themselves. The activity was purposely created to set up an understanding of biconditionals. If in Lesson 4 it was established that not all converse, inverse, and contrapositive statements of a conditional are true then students may understand the point more. If this was not done then consider using a few examples from Lesson 4 before releasing students to do the activity.

If time permits, consider asking students to include the actual statement with each type. This can be based on data from a previous lesson. For example, if you noticed most students struggled with inverse, have them write the inverse statement as part of this activity for each question.

2.5.2 Activity: Conditionals and Truth Values (continued)

4. **Conditional**: If a line or ray divides an angle into two equal angles, then the line or ray is the bisector of the angle.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

5. **Conditional**: If lines that lie on the same plane do not intersect, then the lines are parallel.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

6. **Conditional**: If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

7. **Conditional**: If $\overrightarrow{XY}$ is in the interior of $\angle WXZ$, then $m\angle WXY + m\angle YXZ = m\angle WXZ$.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

Activity Facilitation Tip:

- *The answers to these truth-value questions are all "true." As you are circulating the room, you can determine if a student has a misconception by seeing the word "false."*
- *When addressing potential misconceptions with students, first ask them to tell you how the statement in question should be written. If they applied the wrong definition to it, then the misconception is with understanding converse, inverse, or contrapositive. If students have the statement correct, their misconception may be with the mathematical concept that is being described in the conditional statement.*

For question 7, students may not be familiar with the vocabulary "interior of an angle." This may need to be clarified prior to this question as the area between the two rays that define the angle.

8. **Conditional:** If two planes intersect, then their intersection is exactly one line.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

9. **Conditional:** If two lines have the same slope, then they are parallel.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

For question 8, students may not be familiar with intersecting planes. A quick discussion and demonstration of parallel and intersecting planes may be needed prior to this question.

For question 9, students will have to recall their knowledge of the equations of parallel lines from Algebra 1. Students may have forgotten what is true about the slopes of parallel lines. Consider a brief reminder/refresher on what is true about the slopes of parallel lines if needed.

2.5.3 Bring It Together: Biconditionals

Component Overview: *Students will take what they observed in the prior activity and bring it together to learn about the formal definition of biconditional statements. This activity can be done with a partner or small group but will need whole-group summary for a wrap-up. If needed, you can also deliver this through a Socratic seminar using the questions in the activity.*

2.5.3 Bring It Together: Biconditionals

1. How many of the conditionals from the previous Activity were considered true for all of the statements — converse, inverse, and contrapositive?
   All of the conditionals were considered true for all of the statements.

2. For how many conditionals was the converse of the conditional considered true?
   All of the conditionals were considered true for all of the converse statements.

A conditional statement is considered ***biconditional*** when both the statement and its converse are true. In other words, if both directions of the statement are true – "If P, then Q" and "If Q, then P" – the statement is biconditional.

Biconditional statements can be written in the format of "**if and only if.**"

A **definition** is an exact statement, written formally, of the meaning or description of a word. Even though definitions in mathematics are often written as conditional statements, all definitions are actually biconditional statements.

3. Write each conditional using "if and only if."
   a. If the sum of two angles is $90°$, then the angles are complementary.
      The sum of two angles is 90° if and only if the angles are complementary.
   b. If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.
      The opposite sides of a quadrilateral are parallel if and only if the quadrilateral is a parallelogram.

Consider a foldable, Frayer Model, or type of interactive notebook for this summary. Examples, non-examples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

2.5.4 Your Turn!

Component Overview: *Students will work independently to identify the converse, inverse, and contrapositive statements for a given conditional statement. They will then explain why the conditional statement is biconditional and write the conditional using "if and only if."*

2.5.4 Your Turn!

1. Complete the table for the conditional statement.

If all four angles at the point of intersection of two lines are right angles, then the lines are perpendicular.

Relationship to Conditional	Statement	True or False
Converse	*If two lines are perpendicular, then all four angles at the point of intersection are right angles.*	*True*
Inverse	*If all four angles at the point of intersection of two lines are not right angles, then the lines are not perpendicular.*	*True*
Contrapositive	*If two lines are not perpendicular, then all four angles at the point of intersection of two lines are not right angles.*	*True*

2. Explain why the conditional statement is biconditional.
   The conditional and the converse statement are both true, so the conditional qualifies as a biconditional.

3. Write the conditional using "if and only if."
   All four angles at the point of intersection of two lines are right angles if and only if the lines are perpendicular.

Use the data from the prior two activities to determine which students may need additional support. Some areas of support may be:

- *Writing converse statements*
- *Writing inverse statements*
- *Writing contrapositive statements*
- *Determining truth values for statements*
- *Understanding the conditions necessary to classify a statement as biconditional*

Do you think that a conditional statement can be biconditional if the converse is true but the contrapositive is not? Why or why not?

2.5.5 Wrap-Up: Reflection

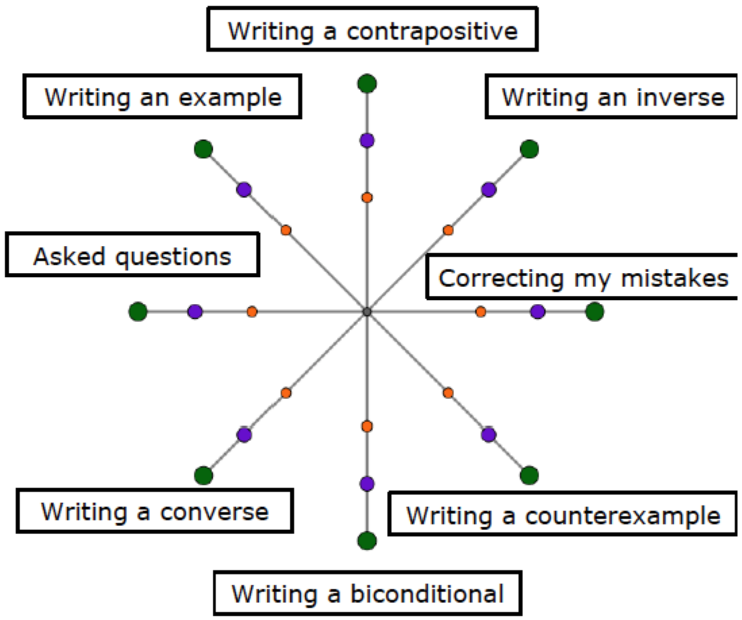
Component Overview: *Students will complete a self-reflection on their level of confidence with eight components of this lesson. This activity should be done independently to get accurate student-level data on their perception of their own confidence related to this content.*

2.5.5 Wrap-Up: Reflection

1. Place an **X** on the circle that shows what you learned or completed during today's lesson.

Answers will vary.

● (green)	I am confident I know how to do this!	OR	I always did this!
● (purple)	I am fairly sure I know how to do this.	OR	I tried this.
● (orange)	I need support on this.	OR	I had trouble with this.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.